AE5188 Response to review 2026/09/07		

1	A more serious objection arises in the n=5 and 6 cases where transformations are not sampled
	exhaustively. The Hermite normal forms are a nice way of enumerating the supercell
	transformations, but they are a bad set of matrices for the given application where they will be
	composed with the 3480 unimodular matrices, because the HNFs often have bigger elements
	than necessary.
	For example, we can take the index-5 supercell pair of 10,10,10,90,90,90 and
	10,22.4,22.4,90,90,90 where b and c are larger by sqrt(5). A simple cell transformation would
	be b'=2b-c, c'=b+2c, thus the transformation we wish to discover is M = [1 0 0 / 0 2 -1 / 0 1 2].
	The corresponding HNF is H = [1 0 0 / 0 5 0 / 0 3 1], which can easily be verified to generate
	the same points as M. By the manuscript's current approach, we write M = UH or M = HU and
	hope that U is one of the 3480 small-entry-unimodular basis transformations. We can easily
	enumerate the 31 HNFs and compute M*H^-1 and H^-1*M to verify that there is no U with the
	required small integer entries. The following Python code reproduces this demonstration:
	import numpy as np
	M = np.array([[1, 0, 0],
	[0, 2, -1],
	[0, 1, 2]])
	hnfs = []
	for a, d, f in [(5, 1, 1), (1, 5, 1), (1, 1, 5)]:
	for b in range(a):
	for c in range(a):
	for e in range(d):
	hnfs.append(np.array([[a, 0, 0],
	[b, d, 0],
	[c, e, f]]))
	assert len(hnfs) == 31
	for H in hnfs:
	U = np.linalg.inv(H) @ M
	integral = np.allclose(U, np.round(U))
	if integral: print(U)
		a bad set of matrices for the given application where they will be composed with the 3480 unimodular matrices,
		
		The reviewer is correct in that the set of matrices is not optimal. The complete set for 5 and 6 is very large (although concurrency might help), so a restricted set has been constructed. If the reviewer can contribute a better, more complete but accessible set, then we would be glad to include it.

2	Supercells correspond to sublattices, no? The title "Approximate Supercell (Superlattice) Matching" doesn't sound right to me for that reason.	**Thank you, I removed superlattice from title and the text**
3	Table 1: "For n = 3 and n = 4 the complete set is used with the 13 and 35 Hermite normal forms appended, respectively; those matrices are already present in the complete set, so the appended copies are redundant and are retained only as a guard." Why?? It's harmless but strikes the reader as very strange to include these vestigial transformations.	**removed the n=3, n=4 description. The reviewer is correct.**

4	Table 1 again: "Left multiplication preserves the lattice and supplies alternate settings of it; right multiplication changes the lattice." The last clause is very mystifying, I made sense of it eventually but I wish it were explained better. "The lattice" here is H*B, a sublattice corresponding to a particular HNF. If we insert the unimodular resetting matrix U, then yes H*U*B is a different lattice. However it was a painful effort to puzzle out that it's different relative to _what_, so I strongly recommend the authors revisit this to try and add some clarity.	**The text has been corrected. What it was supposed to say was that the H and B matrices are multipled in both orders (AB and BA). But that was also not needed and it has been removed.**
5	Appendix: I really don't think the code excerpt here is necessary, we can all surmise that it's possible to efficiently compute all the matrices with these entries and this determinant.	**deleted as requrested by the reviewer**